

EVALUATING MULTILINGUAL RELIABILITY OF LARGE LANGUAGE MODELS IN SCIENTIFIC LITERATURE ANALYSIS

Abstract

Large Language Models (LLMs) are increasingly being used to search, summarize, synthesize, classify, and reason over scientific literature. Their ability to process natural language across multiple languages creates an opportunity to broaden access to scientific knowledge beyond English-dominant research ecosystems. However, multilingual capability should not be equated with multilingual reliability. A model may produce fluent text in a target language while incorrectly translating scientific terminology, omitting qualifications, hallucinating evidence, citing irrelevant sources, or producing conclusions that differ from those obtained in another language. These problems become particularly consequential in scientific literature analysis, where outputs must preserve technical meaning, empirical uncertainty, numerical information, causal relationships, and evidence provenance.

This paper reviews the emerging literature on multilingual large language models, scientific language models, multilingual evaluation benchmarks, factuality assessment, retrieval-augmented generation, citation verification, and scientific question answering. It develops a multidimensional framework for evaluating multilingual reliability in scientific literature analysis based on semantic fidelity, factual correctness, evidence grounding, citation validity, cross-lingual consistency, robustness, calibration, and reproducibility. Existing developments including multilingual foundation models, multilingual benchmarks, scientific retrieval systems, citation-aware generation, and human-centered evaluation are critically compared. Recent benchmarks such as MEGEVERSE, Global MMLU, MMLU-ProX, ScholarBench, and multilingual fact-checking datasets illustrate progress toward broader language coverage, while scientific resources such as SciFact, PubMedQA, LitSearch, PaperQA, S2ORC, and SciReviewGen demonstrate the growing sophistication of literature-oriented evaluation.

The review identifies a major research gap: existing evaluations generally assess either multilingual language ability or scientific reasoning, but rarely evaluate both dimensions jointly under evidence-grounded, cross-lingual conditions. A comprehensive evaluation should therefore compare semantically equivalent scientific tasks across languages, test retrieval and citation fidelity, measure disagreement under language switching, and include expert review for terminology and domain-specific claims. Future research should prioritize multilingual scientific benchmarks, provenance-aware metrics, uncertainty calibration, low-resource languages, contamination-resistant evaluation, and human-centered validation.

Keywords: Large Language Models, multilingual NLP, scientific literature analysis, reliability, factuality, hallucination, cross-lingual consistency, retrieval-augmented generation, scientific question answering, citation verification

1. Introduction

Scientific literature is expanding at a rate that makes comprehensive human reading increasingly difficult. Millions of scientific articles, conference papers, preprints, reviews, and technical reports are now available across disciplines, creating a substantial need for computational systems capable of information retrieval, summarization, evidence synthesis, and question answering. Early domain-specific language models such as SciBERT demonstrated that scientific language differs sufficiently from general-domain text to motivate specialized pretraining, while large scientific models such as Galactica explored the possibility of using foundation models as interfaces to scientific knowledge (Beltagy et al., 2019; Taylor et al., 2022).

The emergence of general-purpose LLMs has expanded this capability considerably. Modern models can summarize papers, explain equations, identify evidence, compare methods, answer research questions, and generate literature-review drafts. Systems such as PaperQA explicitly combine retrieval and generation to answer questions over scientific articles, while LitSearch evaluates the retrieval of papers in response to realistic research questions (Lála et al., 2023; Ajith et al., 2024). Citation-aware systems such as ALCE further recognize that useful scientific assistance requires not merely plausible prose but externally verifiable evidence (Gao et al., 2023).

At the same time, scientific communication is multilingual. Although English remains dominant in much of the international scientific literature, researchers and students increasingly interact with AI systems in languages other than English. Multilingual models such as BLOOM and Aya demonstrate that a single model can process dozens or even more than one hundred languages, while benchmarks such as MEGEVERSE evaluate state-of-the-art LLMs over many languages, including low-resource African languages (BigScience Workshop, 2022; Üstün et al., 2024; Ahuja et al., 2024). The central question, therefore, is no longer whether LLMs can generate text in multiple languages. The more important question is whether they preserve scientific reliability across those languages.

Multilingual reliability presents a distinctive challenge because language is not merely a surface-level representation of scientific information. Scientific meaning is encoded in technical terminology, mathematical expressions, quantities, hedging, uncertainty, methodological descriptions, statistical claims, and relations among evidence and conclusions. Translation or cross-lingual generation can distort any of these components. Moreover, multilingual benchmarks themselves may introduce artifacts. Global MMLU, for example, demonstrates that translating English evaluation data can preserve Western-centric assumptions and introduce language-specific distortions; consequently, benchmark rankings can change when questions are evaluated with greater cultural and linguistic sensitivity (Singh et al., 2025).

The reliability problem is further complicated by hallucination. Factuality research has shown that LLMs can generate unsupported information even when their outputs are fluent and persuasive (Min et al., 2023; Manakul et al., 2023). In literature analysis, an unsupported scientific statement can be especially harmful because it may be interpreted as a claim backed by published evidence. Citation errors add another layer of risk: an LLM may generate a citation that exists but does not actually support the statement it accompanies, or synthesize several papers into a conclusion that no individual source establishes (Gao et al., 2023).

This paper therefore investigates multilingual reliability as a multidimensional property of LLM-based scientific literature analysis. It reviews the current literature, identifies limitations in existing evaluation paradigms, compares emerging approaches, and proposes a comprehensive conceptual framework for future evaluation.

2. Background and Related Work

2.1 Multilingual Large Language Models

Multilingual language modeling predates the current generation of LLMs. Models such as mBERT and multilingual sequence-to-sequence architectures established the feasibility of sharing representations across languages. Later work emphasized multilingual transfer, where knowledge acquired in high-resource languages can improve performance in lower-resource languages. XTREME-R formalized this direction through a multilingual benchmark spanning 50 typologically diverse languages and multiple cross-lingual understanding tasks (Ruder et al., 2021).

The development of large multilingual generative models made cross-lingual prompting increasingly practical. BLOOM is a 176-billion-parameter open-access multilingual model trained on the ROOTS corpus, which includes 46 natural languages and 13 programming languages (BigScience Workshop, 2022). Aya extended this direction to 101 languages and explicitly targeted the underrepresentation of lower-resource languages in multilingual instruction tuning. Its evaluation suite covered 99 languages and included both automated and human evaluation (Üstün et al., 2024).

Research increasingly demonstrates that multilingual capability is uneven rather than binary. MEGEVERSE evaluated several contemporary models across 22 datasets and 83 languages and reported substantial variation across languages and tasks, including stronger performance for larger models on several low-resource languages (Ahuja et al., 2024). Such findings support a critical distinction between *language coverage* and *language reliability*: a model's ability to accept and generate a language does not guarantee equivalent reasoning or factual performance in that language.

Recent surveys similarly identify data imbalance, alignment, transfer, cultural bias, and evaluation quality as central problems in multilingual LLM research (Qin et al., 2024; Zhu et al., 2024; Xu et al., 2025).

2.2 Scientific Language Models and Scientific Literature Processing

Scientific language models are motivated by the technical vocabulary, discourse structures, and reasoning patterns found in scientific texts. SciBERT was pretrained on scientific publications and demonstrated gains over general-domain BERT on several scientific NLP tasks, illustrating the value of domain-specific pretraining (Beltagy et al., 2019).

Large scientific corpora such as the Semantic Scholar Open Research Corpus (S2ORC) have provided infrastructure for scientific text mining. S2ORC contains 81.1 million English-language academic papers and structured full text for millions of open-access papers, including bibliographic relationships and citation information (Lo et al., 2020). Its predominantly English

composition, however, illustrates an important limitation for multilingual scientific analysis: the computational infrastructure for scientific NLP is itself disproportionately English-centric.

Galactica represents an early attempt to train a large generative model specifically on scientific knowledge. The model was trained on scientific papers, reference material, knowledge bases, and related data and demonstrated strong results on several scientific reasoning and knowledge tasks (Taylor et al., 2022). However, the challenges associated with factuality and deployment illustrate that larger scientific pretraining alone cannot guarantee reliable evidence synthesis.

Scientific question-answering datasets have also become important. PubMedQA requires reasoning over biomedical research abstracts and asks models to answer yes/no/maybe research questions from evidence contained in scientific text (Jin et al., 2019). SciFact introduced scientific claim verification, where systems must retrieve evidence-containing abstracts and determine whether those abstracts support or refute a scientific claim (Wadden et al., 2020). These resources make an important conceptual transition from generic language understanding to evidence-oriented scientific reasoning.

2.3 Literature Retrieval and Evidence Synthesis

Scientific literature analysis depends on retrieval as much as generation. LitSearch introduced a benchmark of 597 realistic scientific literature-search questions and found substantial performance differences between retrieval systems, demonstrating that finding the relevant paper is itself a complex reasoning problem (Ajith et al., 2024).

PaperQA goes further by retrieving passages from full-text scientific articles, assessing evidence relevance, and generating answers grounded in retrieved literature. Its LitQA benchmark was designed to better approximate the kind of retrieval and synthesis required for real scientific research (Lála et al., 2023).

These approaches motivate retrieval-augmented generation (RAG) as a mechanism for improving factuality and provenance. However, retrieval does not automatically solve reliability problems. An LLM can retrieve relevant papers but misinterpret them, combine contradictory findings incorrectly, or cite evidence that only partially supports the generated claim.

2.4 Factuality, Hallucination and Citation Reliability

LLM hallucination is a central reliability issue. SelfCheckGPT demonstrated that inconsistencies across independently sampled generations can be used as a black-box signal of hallucination without requiring access to the model's probabilities or an external database (Manakul et al., 2023). FActScore proposed a more fine-grained approach by decomposing generated text into atomic factual claims and estimating what proportion are supported by a reliable source (Min et al., 2023).

Citation reliability represents an additional dimension. ALCE introduced automatic evaluation of citation-supported generation across fluency, correctness, and citation quality. Its findings indicate that even strong systems can generate answers for which citation support is incomplete (Gao et al., 2023). This is particularly relevant to scientific applications because the presence of citations can create a false impression of reliability unless each claim is linked to appropriate evidence.

3. Multilingual Reliability in Scientific Literature Analysis

3.1 Defining Reliability

For scientific literature analysis, multilingual reliability should be understood as the extent to which a model preserves scientifically meaningful information when performing a task across languages.

A useful conceptual definition is:

$$R = f(S, F, E, C, K, U, B, P)$$

where:

- S = semantic fidelity,
- F = factual correctness,
- E = evidence grounding,
- C = citation validity,
- K = cross-lingual consistency,
- U = uncertainty calibration,
- B = robustness to linguistic variation,
- P = provenance and reproducibility.

The key insight is that no single accuracy score adequately captures multilingual scientific reliability.

For example, a model can correctly classify the topic of a paper while mistranslating a numerical result. It can generate a fluent summary while deleting an important limitation. It can identify an appropriate paper but incorrectly attribute the paper's conclusion to a different study. Consequently, conventional language benchmarks and generic QA accuracy provide only partial evidence.

3.2 Semantic Fidelity

Scientific translation must preserve much more than grammatical meaning. It must preserve specialized terms, modality, negation, quantities, units, temporal relations, statistical language, and logical relations.

Consider the difference between:

“The intervention was associated with a reduction in risk”

and

“The intervention reduced risk.”

The first describes an association, whereas the second may imply a stronger causal interpretation. An LLM that simplifies such language while translating or summarizing has introduced a scientific distortion even if the sentence remains linguistically fluent.

Multilingual machine translation benchmarks such as FLORES-101 and FLORES-200 are useful because they enable aligned evaluation across many languages. FLORES-101 contains professionally translated, fully aligned sentences in 101 languages and was explicitly designed to improve evaluation for low-resource languages (Goyal et al., 2022). Later work scaling neural machine translation to 200 languages similarly emphasized the unequal availability of training data and the importance of reducing performance disparities across languages (NLLB Team, 2024).

For scientific analysis, however, general translation benchmarks are insufficient. Scientific evaluation requires domain-specific terminology, experimental descriptions, equations, statistical findings, and methodological caveats.

3.3 Factual Consistency

Scientific reliability requires that a model's claims be supported by the literature it analyzes. This creates at least two factuality settings.

Closed-book factuality measures whether the model produces scientifically correct information without external evidence.

Evidence-grounded factuality measures whether the model's output is entailed by the retrieved source material.

The latter is generally more appropriate for literature analysis because the objective is not simply to reproduce information stored in model parameters but to analyze the contemporary evidence base.

SciFact demonstrates the value of explicit claim-evidence evaluation, while FActScore provides a mechanism for evaluating long-form outputs at the level of individual factual statements (Wadden et al., 2020; Min et al., 2023).

In multilingual settings, factuality should be evaluated in at least three configurations:

1. Source and query in the same language.
2. Source and query in different languages.
3. Source translated into another language before analysis.

Comparing these configurations can reveal whether factual performance depends on the language in which information is represented.

3.4 Cross-Lingual Consistency

One of the strongest tests of multilingual reliability is *language invariance*. Suppose a scientific question q is expressed in English and translated into Hindi, Spanish, Arabic, Korean, or another language. Let the model's corresponding answers be $a_1, a_2, \dots, a_n$.

A highly reliable multilingual system should satisfy:

$$\text{Semantics}(a_i) \approx \text{Semantics}(a_j)$$

for equivalent questions, except where cultural or linguistic differences legitimately affect interpretation.

Cross-lingual consistency can therefore be measured by comparing:

- final answers,
- extracted claims,
- numerical values,
- cited papers,
- cited passages,
- confidence estimates,
- and identified limitations.

This criterion is more informative than comparing surface-form similarity. Two outputs may use different wording yet agree scientifically, whereas nearly identical text can still contain a subtle numerical or causal error.

Recent benchmarks reinforce the need for such paired evaluation. MMLU-ProX provides identical questions across 29 languages specifically to facilitate direct cross-lingual comparisons (2025). Global MMLU extends multilingual evaluation to 42 languages while explicitly separating culturally sensitive from culturally agnostic questions (Singh et al., 2025).

3.5 Citation and Evidence Fidelity

Scientific literature analysis frequently produces statements such as:

“Study X found that intervention Y reduced outcome Z.”

A reliable system should satisfy two independent criteria:

Citation Correctness = Is the cited source relevant?

Citation Entailment = Does the cited source support the exact claim?

These criteria are not equivalent. A paper about the same topic may be relevant but may not support the particular conclusion generated.

Multilingual settings introduce another failure mode. The model may retrieve a paper correctly but generate a citation in the target language that changes the interpretation of the original evidence. Cross-lingual paraphrasing can also remove qualifiers such as “may,” “suggests,” “limited evidence,” or “not statistically significant.”

Accordingly, citation evaluation should be performed against the original scientific document rather than against its machine-translated representation alone.

3.6 Calibration and Uncertainty

Scientific literature frequently contains disagreement, limited evidence, null results, and uncertainty. An LLM that produces categorical answers despite ambiguous literature is unreliable even when the answer sounds authoritative.

Calibration concerns whether stated confidence corresponds to actual correctness. A model should ideally be more uncertain when:

- evidence is sparse,
- papers disagree,
- the requested language is low-resource,
- retrieved documents are weakly relevant,
- or the model cannot establish citation support.

This requirement is particularly important because multilingual fluency can mask uncertainty. A well-written answer in a user's native language can appear more authoritative than an equivalent answer in English even when the model's underlying evidence is weaker.

4. Current Approaches and Recent Developments

4.1 Multilingual Foundation Models

Current multilingual approaches largely fall into three categories: multilingual pretraining, multilingual instruction tuning, and cross-lingual transfer.

BLOOM demonstrated that very large multilingual models could be trained openly across dozens of languages. Aya subsequently emphasized broader coverage, particularly for lower-resource languages (BigScience Workshop, 2022; Üstün et al., 2024).

However, scale alone does not eliminate language disparities. MEGEVERSE found substantial variation between languages and tasks, while research on multilingual translation has shown that low-resource languages remain affected by uneven training data and transfer limitations (Ahuja et al., 2024; NLLB Team, 2024).

4.2 Translation-Based Evaluation

A common practical strategy is to translate an English benchmark into multiple languages. This is inexpensive and permits direct comparison with an English baseline.

The approach has major limitations. Global MMLU shows that translation can introduce linguistic artifacts and preserve culturally specific assumptions from English-language datasets. Translation therefore cannot be assumed to create a truly multilingual benchmark (Singh et al., 2025).

TurkishMMLU makes the same methodological point from another direction: locally authored evaluation questions can provide a better representation of culturally and linguistically meaningful knowledge than automatically translated benchmarks (Yüksel et al., 2024).

For scientific applications, benchmark creation should therefore combine professional or expert translation with native-speaker review and, where feasible, original question authoring.

4.3 Human-Centered Evaluation

Automatic metrics remain useful but are insufficient for scientific multilingual evaluation. Human evaluation is especially important when judging:

- translation of technical terminology,
- preservation of uncertainty,
- logical coherence,
- appropriateness of citations,
- cultural or disciplinary assumptions,
- and subtle scientific inaccuracies.

Aya included human evaluation as part of its multilingual assessment framework (Üstün et al., 2024). Global MMLU similarly used compensated professional and community annotators to improve translation and cultural validity (Singh et al., 2025).

ScholarBench provides another useful development by creating a bilingual academic benchmark covering eight research domains and both English and Korean. Its design demonstrates the value of domain-specific multilingual evaluation rather than relying exclusively on generic language tasks (Noh et al., 2025).

4.4 Retrieval-Augmented Generation

RAG is increasingly attractive for scientific applications because it allows models to ground answers in external documents. PaperQA and related systems show that retrieval can enable substantially more evidence-oriented scientific question answering than purely parametric generation (Lála et al., 2023).

A multilingual scientific RAG pipeline can operate in several ways:

1. Retrieve documents in the query language.
2. Retrieve English scientific literature and answer in another language.
3. Translate the query into multiple languages before retrieval.
4. Perform multilingual retrieval followed by cross-lingual evidence synthesis.

The second and third strategies may increase coverage, but they introduce translation-induced retrieval errors. A scientific concept may be translated into a term that does not correspond to the terminology used in the literature, reducing recall.

Consequently, future multilingual RAG evaluation should separately measure retrieval recall, evidence selection, reasoning over evidence, and final-language generation.

4.5 Long-Context Scientific Analysis

Long-context models provide a potential solution for processing complete papers or multiple articles. However, longer context windows do not guarantee that models use all information reliably. Liu et al. demonstrated that LLM performance can deteriorate substantially when

relevant information is located in the middle of a long context, producing a characteristic “lost in the middle” effect (Liu et al., 2024).

This matters for scientific reviews because critical evidence may be buried within methods, results, supplementary text, or limitations sections. The problem may be amplified when multiple languages are present in the same context, such as an English article combined with a translated abstract or multilingual search results.

4.6 Contamination-Resistant Evaluation

Benchmark contamination is another growing concern. MEGEVERSE reported evidence that some models may have encountered multilingual evaluation data during training, potentially inflating performance estimates (Ahuja et al., 2024). More recent work directly investigating FLORES contamination has shown that contamination can influence even apparently unseen translation directions, demonstrating the importance of contamination-aware multilingual evaluation (Tan et al., 2026).

For scientific literature analysis, contamination can be especially difficult to detect because test documents may already be publicly available online. Evaluation should therefore favor newly authored questions, temporally held-out papers, perturbation-based tests, and hidden evaluation sets wherever possible.

5. Comparative Evaluation Framework

A comprehensive multilingual scientific-reliability evaluation should combine the dimensions summarized below.

Dimension	Core Question	Example Evaluation
Semantic fidelity	Is the scientific meaning preserved?	Expert comparison of source and generated text
Factual correctness	Are claims scientifically correct?	Expert judgment / evidence verification
Evidence grounding	Is each claim supported by retrieved evidence?	Claim–passage entailment
Citation validity	Do citations actually support the claims?	Citation correctness and completeness
Cross-lingual consistency	Does equivalent input produce equivalent conclusions?	Parallel prompts across languages
Numerical fidelity	Are values, units and statistics preserved?	Automated extraction + expert checking
Terminological accuracy	Are domain-specific terms translated correctly?	Expert terminology assessment

Dimension	Core Question	Example Evaluation
Calibration	Does confidence reflect reliability?	Expected calibration error / confidence analysis
Robustness	Does performance persist under Perturbation and adversarial paraphrasing and language switching?	tests
Reproducibility	Can another researcher reconstruct the Source identifiers, passages evidence path?	and retrieval logs

A suitable benchmark should contain scientific articles and questions in multiple domains such as medicine, computer science, physics, chemistry, and social science. Each test item should have a high-quality source document, a gold-standard evidence set, and, where possible, multiple professionally reviewed language versions.

6. Research Challenges and Limitations

6.1 English-Centric Scientific Data

The first major challenge is data imbalance. Scientific corpora and resources such as S2ORC remain heavily concentrated in English (Lo et al., 2020). Consequently, a multilingual model may produce an output in a low-resource language while relying internally on English-dominant representations.

This creates a difficult question: is the model truly reasoning in the target language, or is it effectively translating into an internal high-resource representation and then generating an answer? The distinction matters because translation may introduce errors not captured by conventional language benchmarks.

6.2 Domain-Specific Terminology

General multilingual benchmarks rarely represent the specialized vocabulary found in scientific literature. Terms may have multiple translations depending on discipline, region, or research context.

For example, terminology in medicine can differ between clinical and research contexts. Similar problems occur in computer science, chemistry, engineering, and social sciences. A translation that is linguistically correct but scientifically nonstandard can reduce retrieval quality and confuse users.

6.3 Translation Artifacts

Machine translation may introduce “translationese,” alter syntactic structures, or transform culturally specific assumptions. Global MMLU demonstrates that translated benchmarks can distort the intended evaluation objective, while multilingual MT research continues to document substantial differences between language pairs and resource levels (Singh et al., 2025; Goyal et al., 2022).

Scientific text presents an even higher standard because apparently minor changes can alter causal or statistical meaning.

6.4 Hallucinated and Incorrect Citations

Citation fabrication and citation mismatch remain major problems. Even when an LLM produces a genuine paper citation, the referenced study may not actually support the accompanying statement. ALCE's results highlight the difficulty of achieving complete citation support even for capable systems (Gao et al., 2023).

Multilingual citation generation has an additional complication: titles, author names, and technical terminology may be transliterated or translated inconsistently, making verification harder.

6.5 Evaluation Metric Limitations

Automatic metrics such as BLEU and related lexical measures are not reliable indicators of scientific correctness. Two scientifically equivalent statements may differ substantially lexically, while two lexically similar sentences may differ in causal interpretation.

Similarly, generic QA accuracy does not detect omitted caveats or unsupported synthesis. Factuality must therefore be assessed at the claim and evidence levels.

6.6 Expert Evaluation Costs

Expert assessment is expensive and difficult to scale, especially across languages and scientific disciplines. Multilingual scientific evaluation may require both subject-matter experts and native or highly proficient language specialists.

This creates a practical tension between evaluation breadth and evaluation quality. A benchmark with 100 languages but shallow automatic evaluation may be less useful than one covering 20 languages with rigorous expert validation.

6.7 Cultural and Epistemic Bias

Scientific information itself is not culturally neutral in every discipline. Research questions, datasets, assumptions, and terminology can encode regional practices and perspectives. Global MMLU shows how cultural assumptions in benchmark construction can influence multilingual evaluation results (Singh et al., 2025).

For scientific literature, similar concerns apply to disciplines where local practices or population-specific evidence matter, including public health, education, economics, and social science.

7. Research Gaps and Future Directions

The existing literature establishes a strong foundation but reveals a significant gap: multilingual LLM evaluation and scientific LLM evaluation have largely developed as partially separate research areas. Current multilingual benchmarks primarily evaluate general language understanding, reasoning, translation, or cultural knowledge, while scientific benchmarks generally focus on English-language scientific reasoning or retrieval.

The intersection of these dimensions remains comparatively underexplored.

7.1 Multilingual Scientific Benchmarks

A major priority should be the creation of benchmarks specifically designed for scientific literature analysis across languages. Such benchmarks should include parallel or independently authored questions over the same evidence base, professionally validated translations, low-resource languages, and expert-reviewed scientific terminology.

ScholarBench represents a promising direction, but much broader language and disciplinary coverage is required (Noh et al., 2025).

7.2 Language-Symmetric Evaluation

Future benchmarks should evaluate the same scientific task in several languages and compare:

$$\Delta_{lang} = Performance_{best\ language} - Performance_{target\ language}$$

However, the goal should not merely be small score differences. Researchers should compare the *identity of scientific conclusions*, supporting evidence, numerical values, uncertainty, and citations.

A model could receive similar QA accuracy across two languages while producing different interpretations of ambiguous evidence. Claim-level consistency would expose this failure.

7.3 Multilingual Evidence Graphs

A promising future direction is the construction of multilingual evidence graphs in which claims are linked to source passages, papers, authors, datasets, and citations.

For each generated claim c_i , the evaluation system could store:

$$c_i \rightarrow \{p_1, p_2, \dots, p_n\}$$

where each p_j is an evidence passage with document identifier, language, publication metadata, and relevance score.

This would make scientific generation auditable and help distinguish genuine synthesis from unsupported generation.

7.4 Translation-Aware Scientific Retrieval

Future multilingual RAG systems should not assume that the translated form of a query is optimal for retrieval. Instead, retrieval could combine:

- original-language queries,
- translated query variants,
- multilingual embeddings,
- terminology dictionaries,
- domain-specific entity linking,
- and language-specific scientific indexes.

This could reduce errors caused by literal translation of technical terms.

7.5 Low-Resource Scientific Languages

The majority of current research emphasizes major languages. Yet the scientific-reliability question is arguably most important for underrepresented languages, where users have fewer alternative resources for accessing technical information.

Recent developments such as BnMMLU show increasing attention to languages such as Bengali. Its 2026 evaluation covers 41 domains and reports persistent reasoning gaps across evaluated systems, illustrating that language-specific scientific and academic benchmarks can expose weaknesses hidden by broader multilingual evaluations (Joy & Shatabda, 2026).

Future work should extend this approach to Indian, African, Southeast Asian, and Indigenous languages, particularly for STEM and biomedical domains.

7.6 Contamination-Resistant Scientific Evaluation

Evaluation sets should incorporate newly published or withheld scientific literature and questions that are unavailable during model training. Temporal splits are particularly valuable because scientific knowledge changes over time.

Researchers should also use adversarial perturbations, citation replacement, entity substitution, and source-side paraphrasing to determine whether models rely on memorization instead of genuine evidence reasoning.

7.7 Calibrated Uncertainty and Abstention

A reliable scientific system should be able to say that evidence is insufficient. Future multilingual models should therefore be evaluated not only on answer accuracy but also on the appropriateness of abstention.

A useful objective is:

$$Utility = Accuracy - \lambda(Hallucination) + \gamma(Calibrated\ Abstention)$$

where λ reflects the cost of unsupported scientific claims.

This is especially important when the model processes low-resource languages or weakly represented scientific topics.

7.8 Human-in-the-Loop Scientific Analysis

The most realistic future paradigm is unlikely to be completely autonomous literature review. Instead, LLMs may function as research assistants whose outputs remain linked to evidence and subject to expert verification.

A robust workflow could involve:

retrieval → evidence extraction → multilingual synthesis → claim verification → citation verification → expert review

Such systems would shift the focus from generating persuasive text to constructing transparent chains of evidence.

8. Conclusion

Multilingual LLMs offer substantial potential for transforming scientific literature analysis by making research knowledge more accessible across linguistic boundaries. However, multilingual fluency is not equivalent to multilingual reliability. Scientific analysis requires preservation of technical meaning, factual accuracy, evidence grounding, citation correctness, uncertainty, and methodological nuance.

The literature reviewed in this paper shows progress on several fronts. Multilingual foundation models such as BLOOM and Aya have substantially expanded language coverage. Benchmarks including XTREME-R, MEGEVERSE, Global MMLU, MMLU-ProX, and ScholarBench have strengthened multilingual evaluation. Scientific resources such as SciBERT, S2ORC, PubMedQA, SciFact, PaperQA, LitSearch, and SciReviewGen have established increasingly realistic scientific NLP tasks. Meanwhile, FActScore, SelfCheckGPT, and ALCE demonstrate that factuality, hallucination, and citation support must be evaluated explicitly rather than inferred from fluency.

Nevertheless, current benchmarks rarely test the full intersection of multilingualism and scientific evidence reasoning. This constitutes the principal research gap. A genuinely reliable multilingual scientific LLM should be able to analyze equivalent evidence across languages, preserve numerical and terminological information, retrieve relevant sources, produce citations that entail its claims, communicate uncertainty appropriately, and remain consistent when the input language changes.

Future research should therefore move toward language-symmetric, evidence-grounded, contamination-resistant scientific benchmarks with strong human validation. Particular attention should be given to low-resource languages, domain-specific terminology, multilingual retrieval, citation provenance, cross-lingual consistency, and calibrated abstention.

Ultimately, the relevant question is not whether an LLM can *speak* many languages. It is whether scientific knowledge remains accurate, traceable, and epistemically responsible when the language of interaction changes. Establishing that standard is essential before multilingual LLMs can be considered dependable tools for scientific literature analysis.

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